

## Signal Processing

### Exercise 10 – IIR, Correlation, Linear / Circular Convolution

SoSe 2026

#### IIR Filter Design

##### Question 5

An IIR low-pass filter is required to have the following specifications:

- Pass-band edge frequency  $\omega_{\text{pass}} = 0.24\pi$  rad/sample
- Stop-band edge frequency  $\omega_{\text{stop}} = 0.5\pi$  rad/sample
- Maximum pass-band attenuation  $\delta_{\text{pass}} \leq 1$  dB
- Minimum stop-band attenuation  $\delta_{\text{stop}} \geq 4$  dB

Consider a butterworth low-pass analog filter to approximate the filter characteristic. Use the bilinear transform with  $T = 2$  to design the filter. In addition, we would like to make sure that the maximum passband attenuation is met exactly.

The magnitude-squared approximation of a low-pass  $N$ th-order Butterworth filter reads as

$$|H_c(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}. \quad (1)$$

or

$$|H_c(s)|^2 = \frac{1}{1 + \left(\frac{s}{j\Omega_c}\right)^{2N}} \quad (2)$$

Thus, the order of filter is given by

$$N \geq \frac{1}{2} \frac{\log\left(\frac{1}{|H_c(j\Omega_{\text{pass}})|^2} - 1\right) - \log\left(\frac{1}{|H_c(j\Omega_{\text{stop}})|^2} - 1\right)}{\log \Omega_{\text{pass}} - \log \Omega_{\text{stop}}}. \quad (3)$$

- a) Sketch the tolerance scheme of the filter.
- b) Calculate the minimum order  $N$  as well as the cut-off frequency in such a way that the filter specifications are fulfilled.
- c) Determine a digital filter system function  $H(z)$  by using the bilinear transform with  $T = 2$ . Hint: Determine  $|H_c(s)|^2$  and  $H_c(s)$  first. In order to do so, try to obtain the poles of  $|H_c(s)|^2$ . Calculate the filter coefficients  $b_k$ ;  $a_k$  of the standardized system function as well.

## Correlation

In general, the mutual relationship between two or more sets of data is described via correlation. In signal processing, correlation resembles a measure of the similarity of two signals, e.g., the similarity of signal 1 with respect to signal 2 (and vice versa).

### Two types of correlation

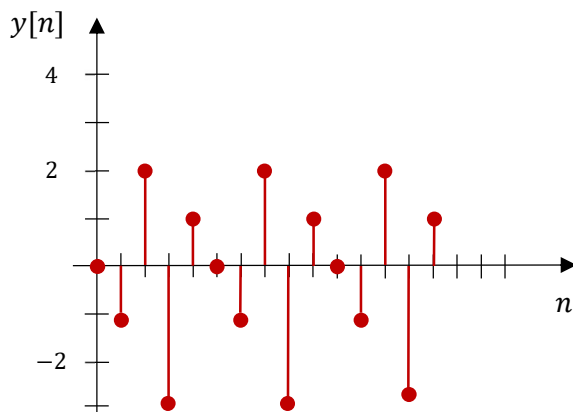
#### Autocorrelation

When we correlate a given signal with itself (time-shifted version of itself), then this correlation type is called autocorrelation.

$$\varphi_{xx}[m] = \sum_{n=-\infty}^{\infty} x[n] \cdot x[n + m]$$

The maximum of the autocorrelated signal  $\varphi_{xx}[m]$  occurs, when the overlapping signal matches the given signal the best → always when time-shift between both is equal to zero.

Application: Verify periodicity of a signal



Maximum sample values are obtained at  $n = 3$ . The next maximum is found at  $n = 6$ . The third maximum is found at  $n = 9$  but it is slightly lower.

Errors are inevitable parts of systems. In this case, the error might be affected due to bit errors.

## Cross-correlation

When we correlate a given signal with another signal to find out how much resemblance exists between them, then this correlation type is called cross-correlation.

$$\varphi_{xy}[m] = \sum_{n=-\infty}^{\infty} x[n] \cdot y[n+m] = \sum_{n=-\infty}^{\infty} x[n-m] \cdot y[n]$$

The maximum of the cross-correlated signal  $\varphi_{xy}[m]$  occurs, when both signals overlap at their best. In this case, the two considered signals become most similar to each other.

Application: Determine signal delays (Radar, Sonar, acoustical distance measurement)

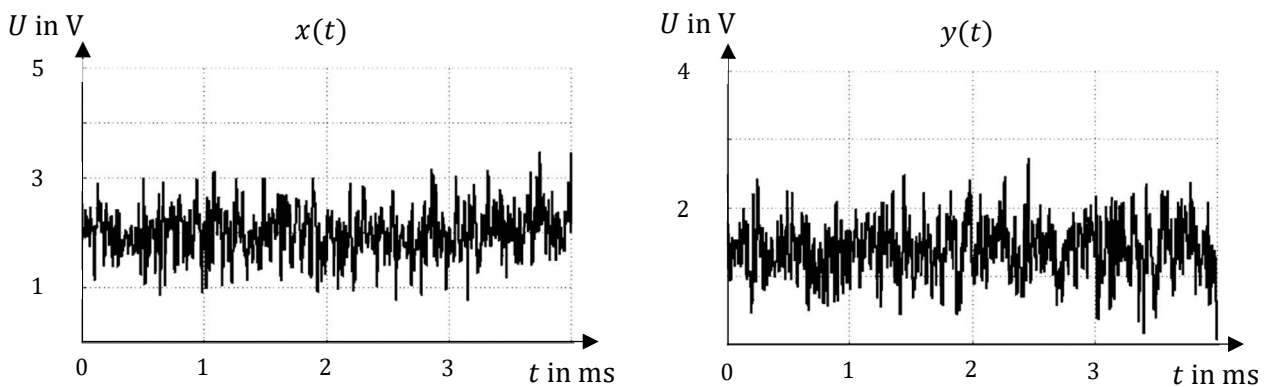


Fig.1: Transmitted signal  $x(t)$  and received signal  $y(t)$ .

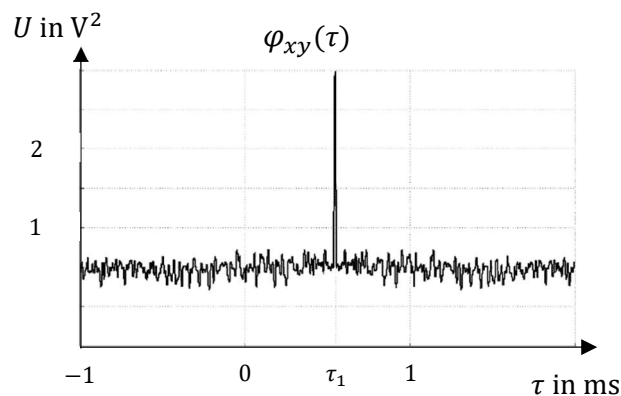
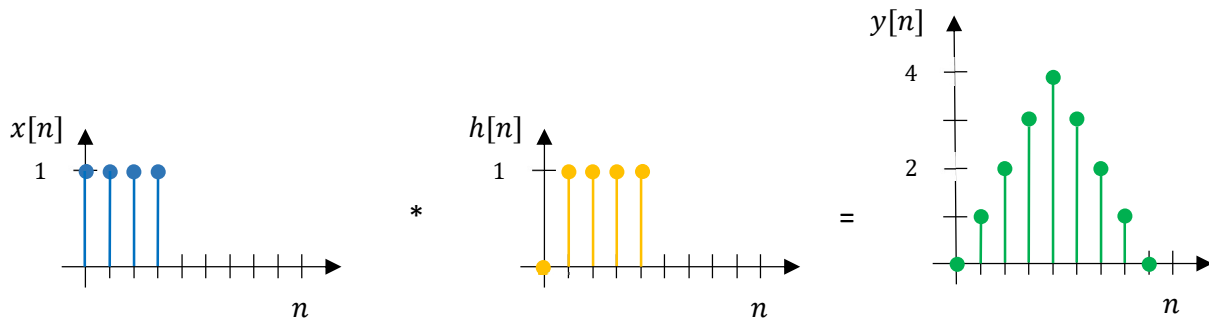


Fig.2: Cross-correlation function  $\varphi_{xy}(\tau)$  as a result of cross-correlating  $x(t)$  with  $y(t)$ .

As depicted in Fig.2, white noise is a very good test signal for time delay measurements using cross-correlation, because there is only one cross-correlation peak at  $\tau_1$ .

## Linear Convolution:

$$y[n] = x[n] * h[n] = \sum_{m=-\infty}^{+\infty} x[m] \cdot h[n - m]$$



When both signals are congruent (100 % overlapping), then we get a maximum value for  $y[n]$ .

## Circular Convolution:

Circular convolution is applied for periodic sequences, whereas linear convolution is employed for aperiodic sequences.

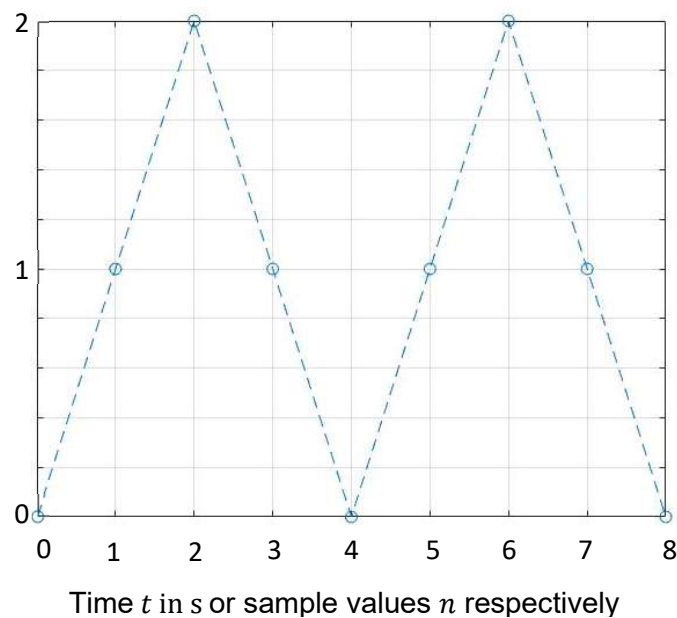
$$y[n] = x[n] \circledast h[n] = \sum_{i=0}^{N-1} x[i] \cdot h[n - i]_N$$

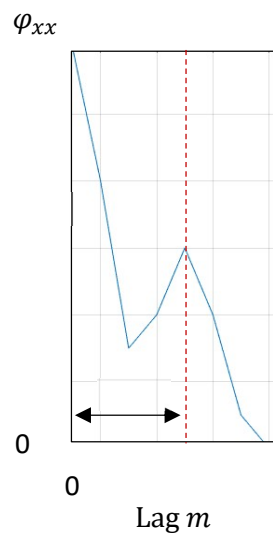
## Question 1

The autocorrelation is applied to determine a certain frequency component of an audio signal. In considering the occurring maxima of the autocorrelation  $\varphi_{xx}[m]$  one can determine the frequency.

Determine the frequency of the given signal  $x[n] = [0, 1, 2, 1, 0, 1, 2, 1, 0]$  by calculating the autocorrelation function  $\varphi_{xx}[m]$ .

$y(t)$  or  $y[n]$  respectively in V



[illegible]

## Question 2

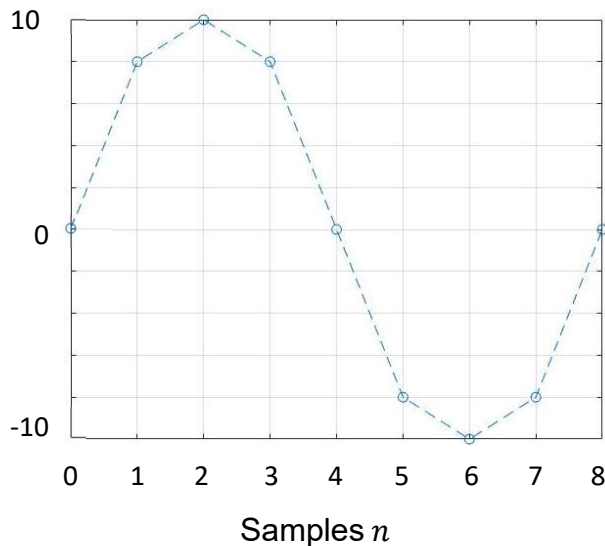
You want to determine the distance from a sound source to an object. Thus, you decide to transmit a single pulse signal. The signal is sampled at  $f_s = 10$  kHz, the experiments are conducted in water (sound velocity  $c_W = 1500 \frac{\text{m}}{\text{s}}$ ).

Apply the cross-correlation and calculate the distance  $d$  from the transmitter to the object.

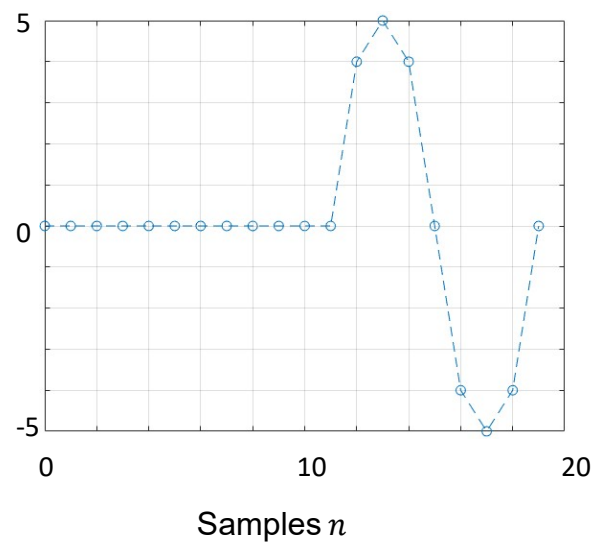
Transmitted signal:  $x[n] = [0, 8, 10, 8, 0, -8, -10, -8, 0]$

Received signal:  $y[n] = [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 4, 5, 4, 0, -4, -5, -4, 0]$

$x[n]$  in V



$y[n]$  in V



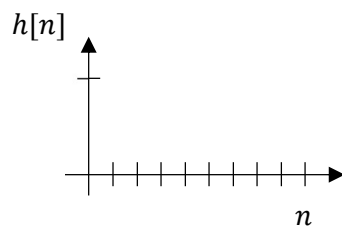
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### Question 3

Your experiments generate a noisy signal  $x[n] = [1, -3, 2, -1, 2]$  of the length  $L = 5$  and you decide to smooth the signal by using a moving average filter (3<sup>rd</sup> order).

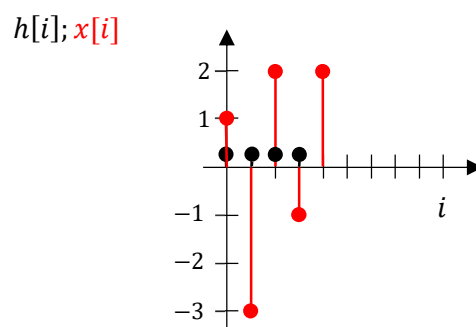
a) Draw the impulse response  $h[n]$ .

3<sup>rd</sup> order  $\rightarrow$  length  $K =$

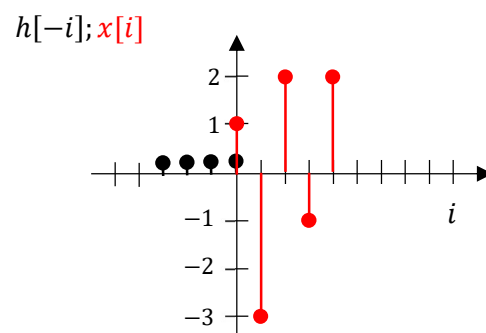


b) Apply the linear convolution to calculate the output signal  $y[n]$ .

1. Plot both signals



2. Mirror impulse response:  $h[i] \rightarrow h[-i]$



3. Shift impulse response  $h[-i]$   $n$ -positions to the right ( $n = 0, 1, \dots, L + K - 1$ ).  
Perform shift by setting  $i = i - n$ .
4. Multiply and sum.

|            | $i \rightarrow$ |  |  | 1 | -3 | 2 | -1 | 2 |  |  |  |  |                |
|------------|-----------------|--|--|---|----|---|----|---|--|--|--|--|----------------|
| $h[0 - i]$ |                 |  |  |   |    |   |    |   |  |  |  |  | $\leftarrow n$ |
| $h[1 - i]$ |                 |  |  |   |    |   |    |   |  |  |  |  |                |
| $h[2 - i]$ |                 |  |  |   |    |   |    |   |  |  |  |  |                |
| $h[3 - i]$ |                 |  |  |   |    |   |    |   |  |  |  |  |                |
| $h[4 - i]$ |                 |  |  |   |    |   |    |   |  |  |  |  |                |
| $h[5 - i]$ |                 |  |  |   |    |   |    |   |  |  |  |  |                |
| $h[6 - i]$ |                 |  |  |   |    |   |    |   |  |  |  |  |                |
| $h[7 - i]$ |                 |  |  |   |    |   |    |   |  |  |  |  |                |

#### Question 4

Calculate the cyclic convolution of the two series  $x[n] = [1, 2, 3, 2, 3, 4]$  and  $h[n] = [1, -1, 2, -2, 1, 0]$  for  $N = 6$ .

|                |   | 1 | 2 | 3 | 2 | 3 | 4 | $y[n]$ |
|----------------|---|---|---|---|---|---|---|--------|
| $\leftarrow n$ | 0 |   |   |   |   |   |   |        |
|                | 1 |   |   |   |   |   |   |        |
|                | 2 |   |   |   |   |   |   |        |
|                | 3 |   |   |   |   |   |   |        |
|                | 4 |   |   |   |   |   |   |        |
|                | 5 |   |   |   |   |   |   |        |